

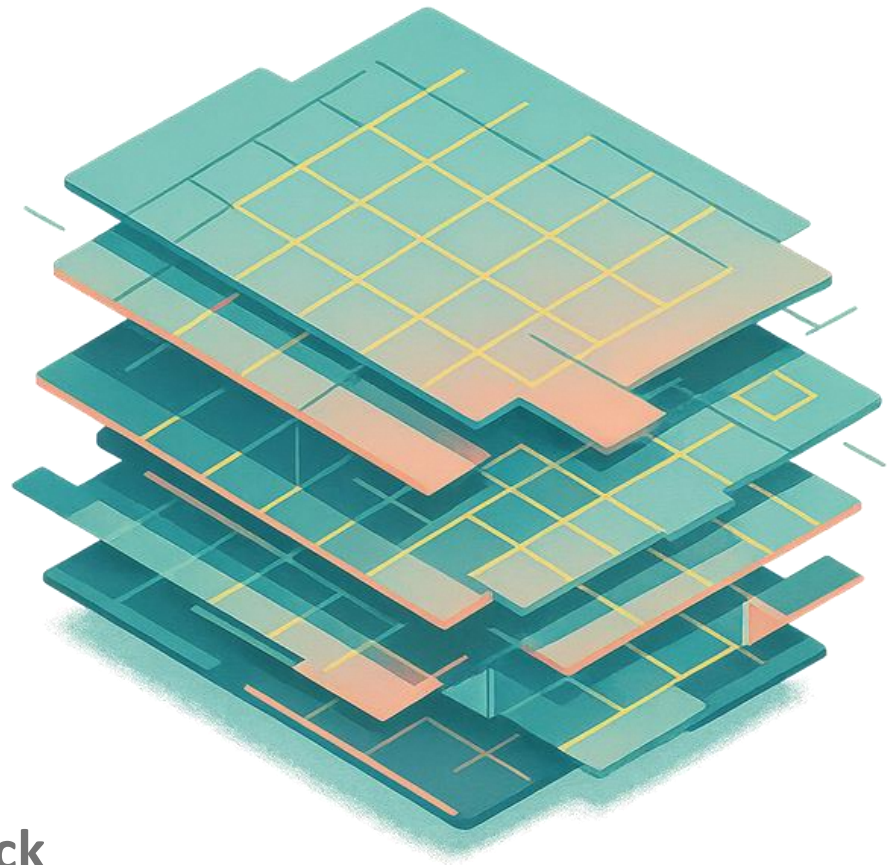


# QuickShop Cloud Data Storage Strategy Proposal

By Div, Cheif Data Officer

# Our Data is Our Most Valuable—and Underutilized—Asset

The Challenge & The Opportunity



## Current State: The Bottleneck



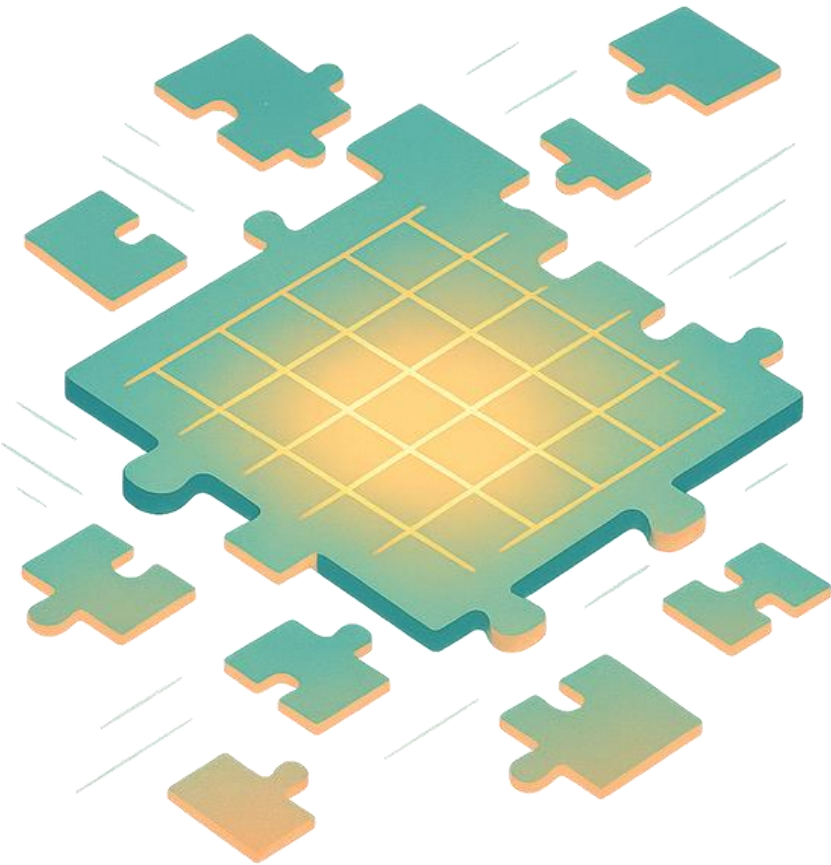
**Slow & Unscalable Systems:** Performance degrades during peak seasons, leading to lost sales and frustrated customers.



**Unpredictable, Rising Costs:** Paying for idle, over-provisioned resources is inefficient and drains the budget.



**Siloed, Fragmented Data:** We lack a single view of our customers, inventory, and operations, hindering personalization and efficiency.



## Future State: The Opportunity



**Real-Time Agility:** Instantly react to market trends, customer behavior, and inventory changes.



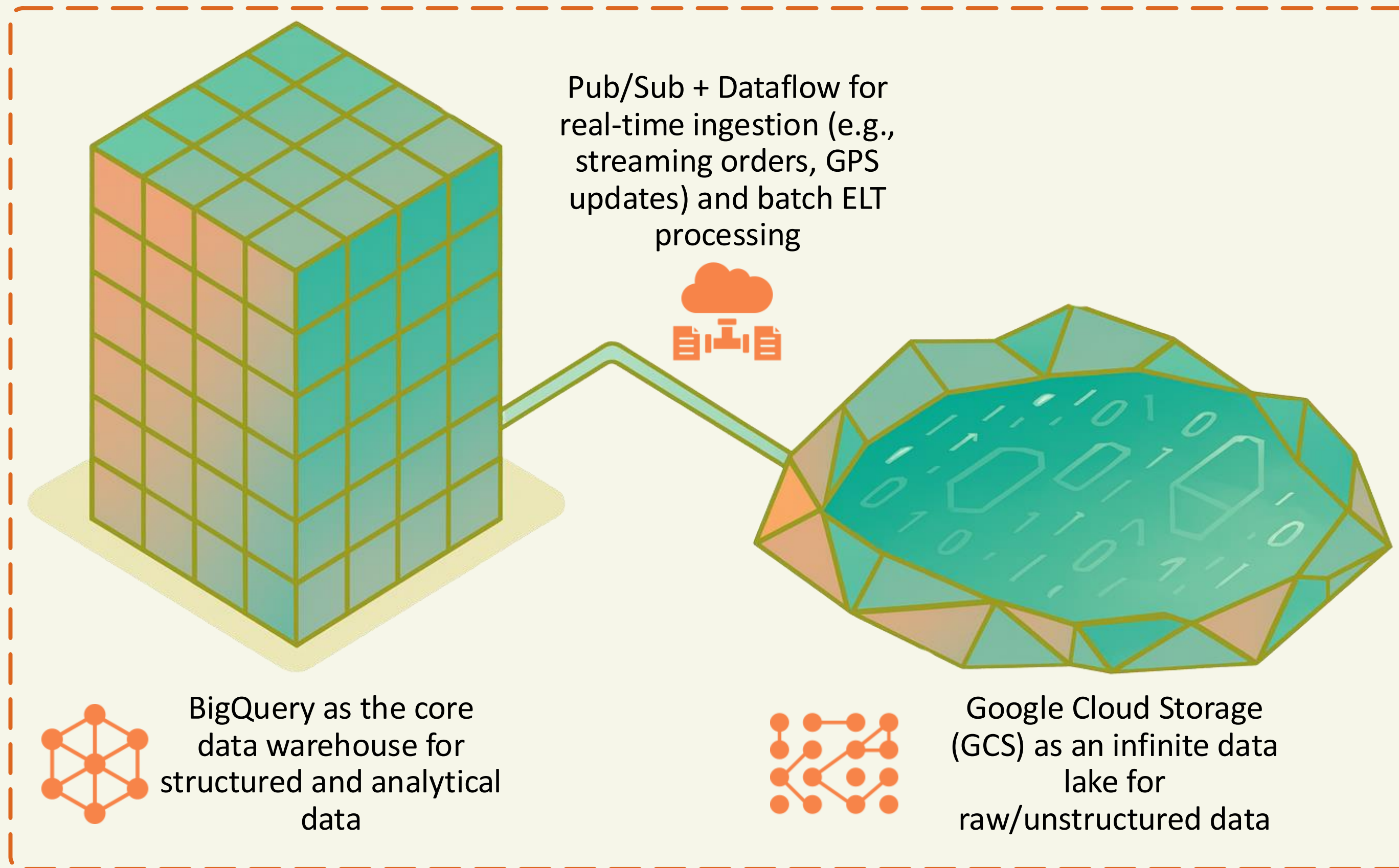
**Optimized, Pay-as-you-go Model:** Cut Total Cost of Ownership (TCO) by 30-40% by paying only for what you use.



**Unified 360° Customer View:** Drive loyalty and revenue with hyper-personalized experiences and targeted marketing.

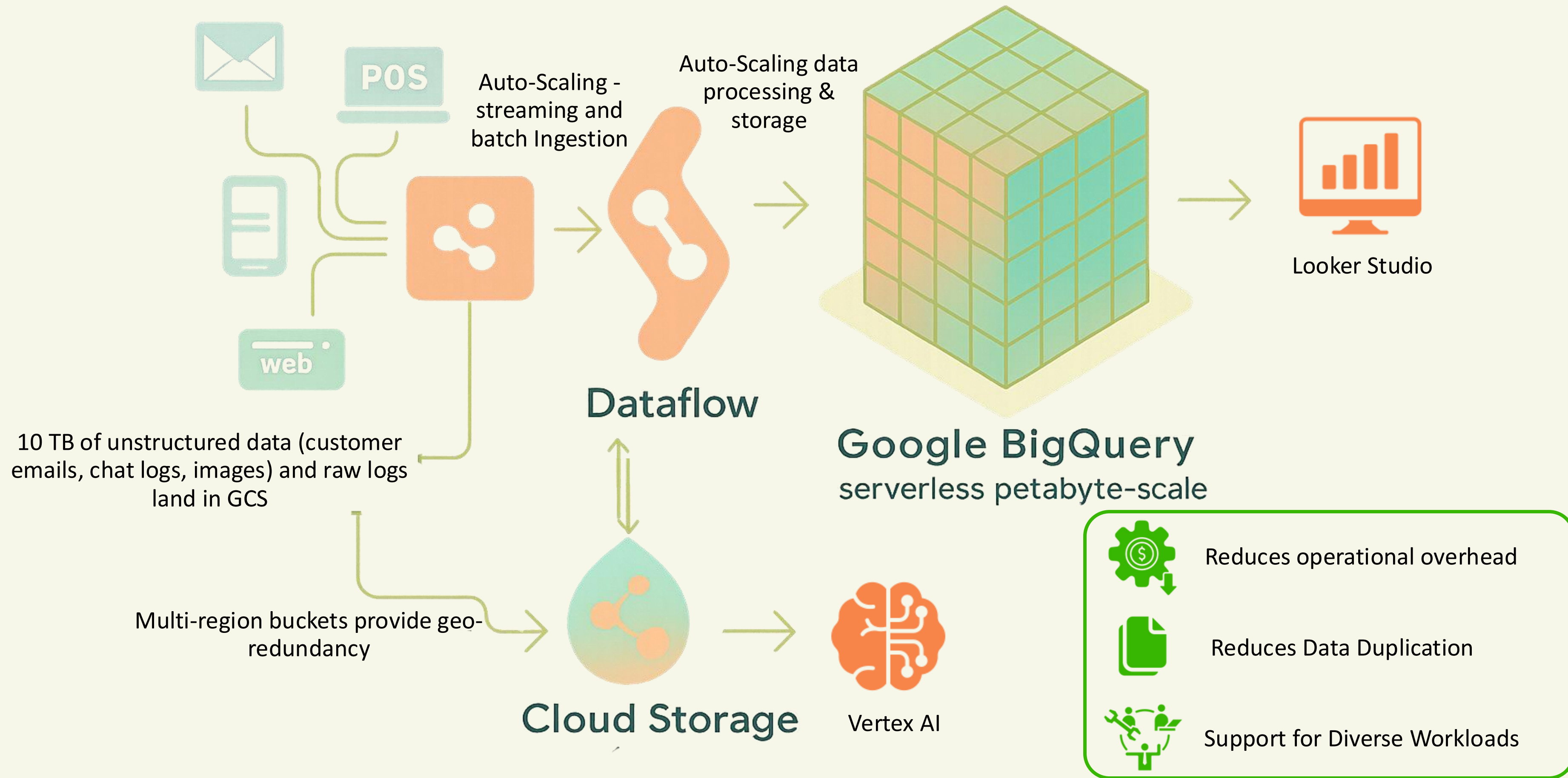


# Proposed Solution (Google Cloud Lakehouse)



Governance: Dataplex plus Data Catalog  
Security: IAM, KMS, Audit logs

# Scalability: Architecture & Storage Design





# Faster Pipelines, More Trusted Data.

The ELT Advantage & The Governance Fabric



## From Slow ETL to Agile ELT

We are shifting from traditional ETL (Extract, Transform, Load) to a modern ELT (Extract, Load, Transform) pattern. This is a fundamental change for speed and simplicity.



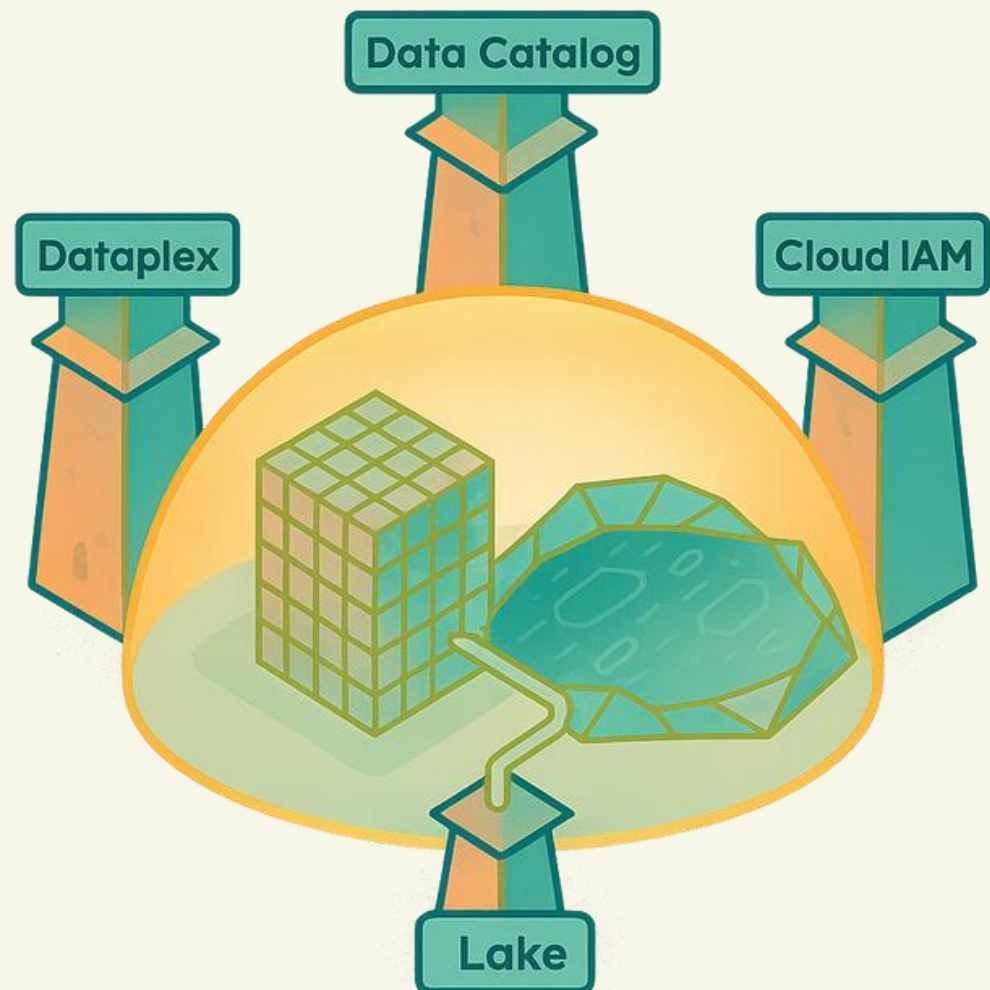
Massively Parallel Queries



Ultimate Flexibility



Democratizes Data



## Unified Governance with Dataplex: The 3 Pillars of Trust

**Dataplex** acts as an intelligent data fabric, supports Apache Iceberg for lakehouse interoperability ensuring our data is reliable and secure

**DISCOVER:** An automated, searchable data catalog with business context and end-to-end lineage.

**SECURE:** robust IAM with role-based access controls and security policies applied consistently across all data.

**TRUST:** Automated data quality checks to monitor freshness, validity, and completeness, quarantining bad data.

# Driving Growth Through Data-Driven Decisions.



**Hyper-personalized marketing**, sophisticated product recommendations, and tailored shopping experiences to boost **customer lifetime value**.



real-time analytics for **dynamic pricing**, **campaign monitoring**, and **demand forecasting**



Automate compliance with **GDPR** and **CCPA**, building customer trust and mitigating regulatory risk



Pay-as-you-go operational model **(OpEx)**. **No** over-provisioning and **No** costly hardware maintenance



Reduce query and reporting times **from hours or days to minutes or seconds**

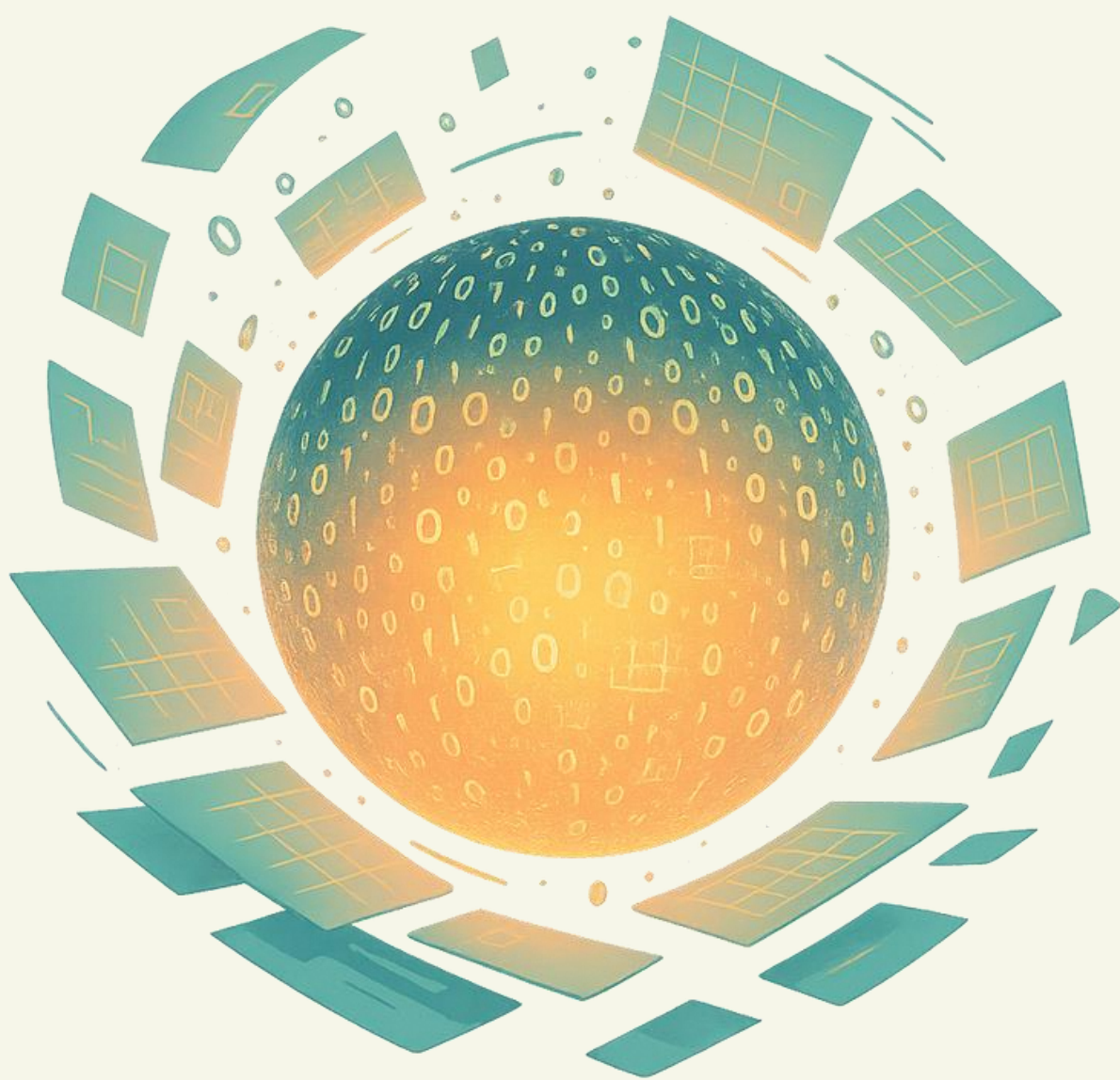


**De - Identification** of Sensitive customer information using DLP by **masking** or **tokenization**

# A Transparent Plan for a Transformative Investment.



Based on our current 5 TB structured & 10 TB unstructured data



| Category              | GCP Services                           | Estimated Monthly Cost (USD)    |
|-----------------------|----------------------------------------|---------------------------------|
| Storage               | BigQuery & Cloud Storage (Tiered)      | $\$100 + \$102 = \$202$         |
| Analysis/Compute      | BigQuery Queries & Dataflow            | $\$625 + \$450 = \$1,075$       |
| Ingestion             | Pub/Sub                                | \$40                            |
| Governance & Security | Dataplex, DLP, Security Command Center | $\$120 + \$150 + \$500 = \$770$ |
| TOTAL ESTIMATE        |                                        | <b>\$2,087 / month</b>          |